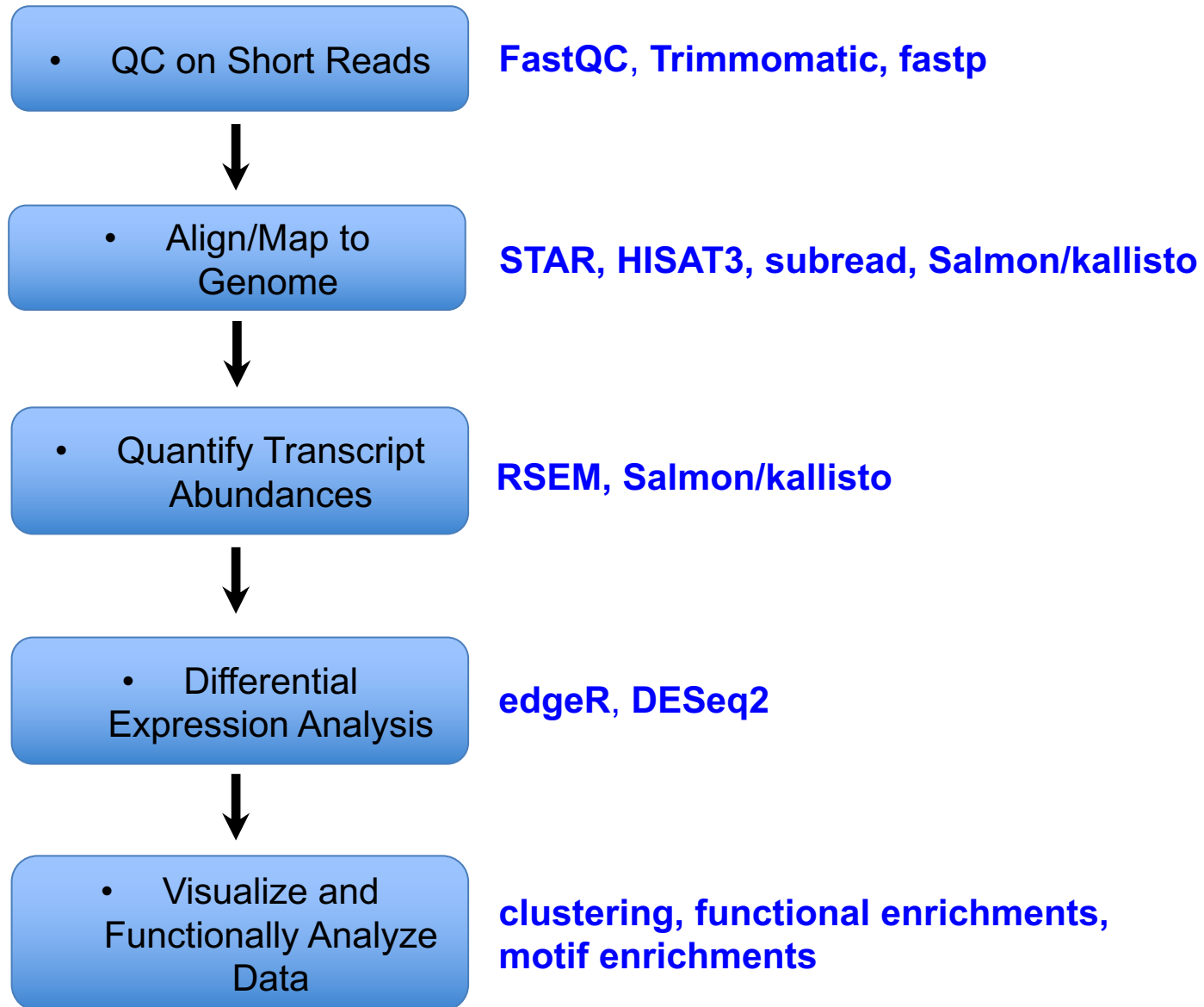


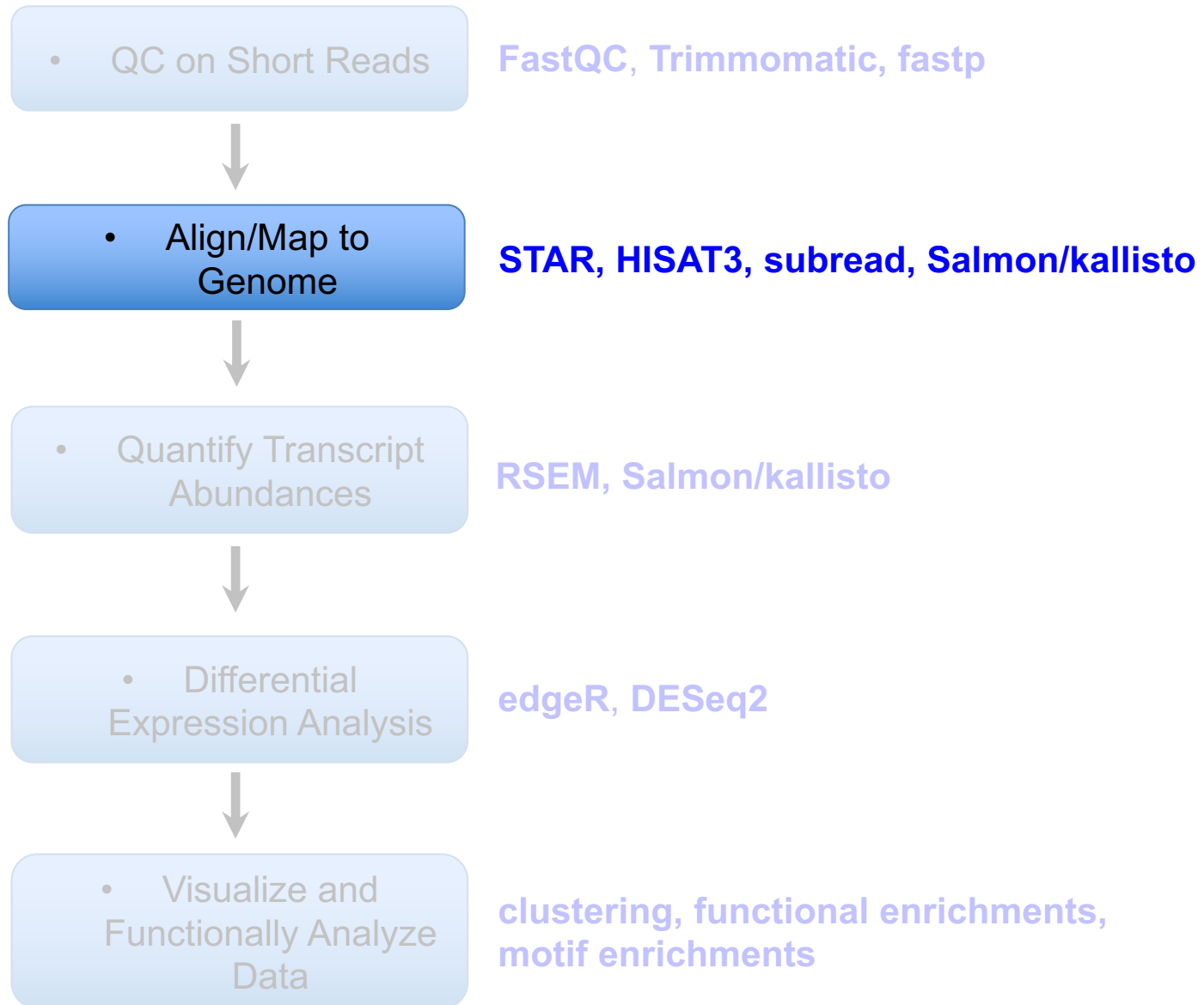
Read Mapping

- Download and open 04_Read_Mapping.qmd and run the first two chunks (to install packages)
- If Rfastp did not work for you last time, download the trimmed files from the OneDrive and save to
~/Desktop/Genomic_Data_Analysis/Data/Trimmed_rfastp
- Tip: If packages do not install automatically, google the package and copy the different installation codes from Bioconductor—one of those should work (and AI is good at troubleshooting)
- For Tuesday, download the Integrated Genome Viewer (IGV):
<https://software.broadinstitute.org/software/igv/download>
 - try Java included versions first, and then Java not included if that version doesn't open correctly

RNA-seq Workflow



RNA-seq Workflow



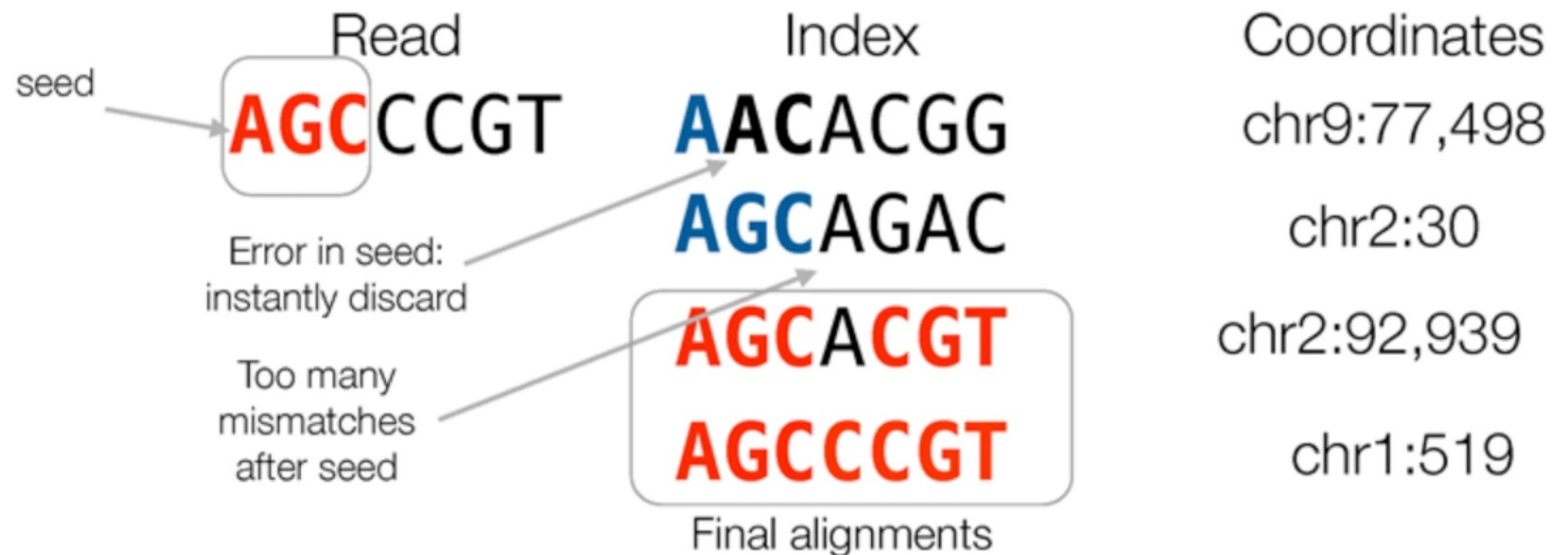
Aligning Reads: the worst possible solution

Naïve, brute force: take a read, and slide it along the genome one nucleotide at a time, finding the best match

- for a human genome with 3 billion bases, that's 3 billion different positions to slide across the genome
- So, for 20 million reads (average depth for RNA-seq), that's 20 million x 3 billion comparisons, or 6 quintillion.
- Now imagine that you also want to allow just one mismatched base

Actual Alignment Strategies

- Most methods use “seed and extend”
- First step is to build an “index,” analogous to the index from a book
- Allows the alignment algorithm to search from DNA strings alphabetically instead of trying to read the whole genome front to back
- Raw reads are compared to index and possible matches are reported



- Harold Pimentel, UC-Berkeley

BWA and Bowtie2

BIOINFORMATICS

ORIGINAL PAPER

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Sequence analysis

Fast and accurate short read alignment with Burrows–Wheeler transform

Heng Li and Richard Durbin*

Wellcome Trust Sanger Institute, Wellcome Trust Genome Campus, Cambridge, CB10 1SA, UK

Received on February 20, 2009; revised on May 6, 2009; accepted on May 12, 2009

Advance Access publication May 18, 2009

Associate Editor: John Quackenbush

Software

Open Access

Ultrafast and memory-efficient alignment of short DNA sequences to the human genome

Ben Langmead, Cole Trapnell, Mihai Pop and Steven L Salzberg

Address: Center for Bioinformatics and Computational Biology, Institute for Advanced Computer Studies, University of Maryland, College Park, MD 20742, USA.

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Published: 4 March 2009

Genome Biology 2009, **10**:R25 (doi:10.1186/gb-2009-10-3-r25)

Received: 21 October 2008

Revised: 19 December 2008

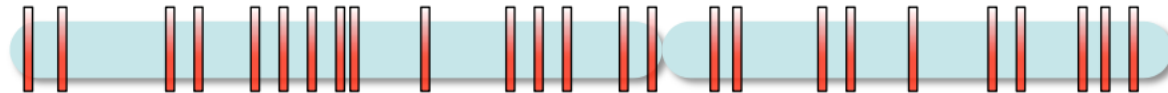
Accepted: 4 March 2009

Bowtie2



```
@HWI-EAS412_4:1:1:1376:380  
AATGATACGGCGACCACCGAGATCTA
```

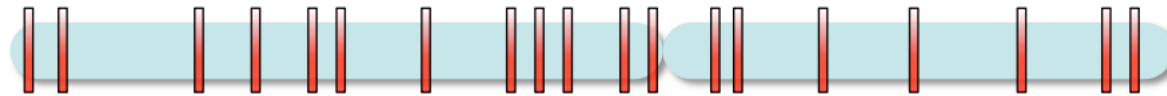
Bowtie2



@HWI-EAS412_4:1:1:1376:380

AATGATACGGCGACCACCGAGATCTA

Bowtie2



@HWI-EAS412_4:1:1:1376:380

AATGATACGGCGACCA**CCGAGATCTA**

Bowtie2



@HWI-EAS412_4:1:1:1376:380
AATGATACGGCGACCACCGAGATCTA

Bowtie2



@HWI-EAS412_4:1:1:1376:380

AATGATACGGCGACCACCGAGATCTA

Bowtie2



@HWI-EAS412_4:1:1:1376:380

AATGATACGGCGACCACCGAGATCTA

Other algorithms speed up the search

- Subread (what we'll use today)
 - uses “seed and vote” algorithm
 - breaks reads into parts (subreads) and then compares against an index of the genome (the genomic location with the most matching subreads for a read “wins”)
- STAR: Spliced Transcripts Alignment to a Reference
 - uses Maximal Mappable Prefix (MMP) to align reads to index
- HISAT: hierarchical indexing for spliced alignment of transcripts
 - faster hierarchical indexing algorithm

Alignment Outputs: SAM and BAM files

- Sequence Alignment Map format
- concise file format that contains mapping information (read sequence, where the reads map to the genome, mapping quality...)

@HD VN:1.5 SO:coordinate															
@SQ SN:ref LN:45															
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*					
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*					
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	*	SA:Z:ref,29,-,6H5M,17,0;				
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*					
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	*	SA:Z:ref,9,+,5S6M,30,1;				
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	*	NM:i:1				

Header
section

Alignment
section

Optional fields in the format of TAG:TYPE:VALUE

QUAL: read quality; * meaning such information is not available

SEQ: read sequence

TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.

PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column.

RNEXT: reference sequence name of the primary alignment of the NEXT read. For paired-end sequencing, NEXT read is the paired read, corresponding to the RNAME column.

CIGAR: summary of alignment, e.g. insertion, deletion

MAPQ: mapping quality

POS: 1-based position

RNAME: reference sequence name, e.g. chromosome/transcript id

FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.

QNAME: query template name, aka. read ID

Alignment Outputs: SAM and BAM files

- Sequence Alignment Map format
- concise file format that contains mapping information (read sequence, where the reads map to the genome, mapping quality)
- mapping quality provided as Phred scores for each alignment (can use as cutoffs in downstream analysis)
- for RNA-seq, is often further compressed into a Binary Alignment Map (BAM) file, which is used to generate counts per gene/transcript

Pseudo-alignment and Quasi-mapping

nature
biotechnology

BRIEF COMMUNICATIONS

Near-optimal probabilistic
RNA-seq quantification

Nicolas L Bray¹, Harold Pimentel², Páll Melsted³
& Lior Pachter^{2,4,5}

NATURE METHODS

BRIEF COMMUNICATIONS

**Salmon provides fast and
bias-aware quantification
of transcript expression**

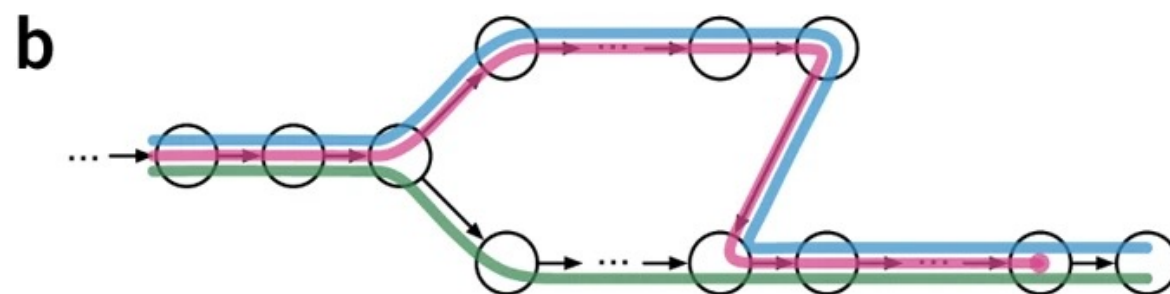
Rob Patro¹, Geet Duggal², Michael I Love^{3,4},
Rafael A Irizarry^{3,4} & Carl Kingsford⁵

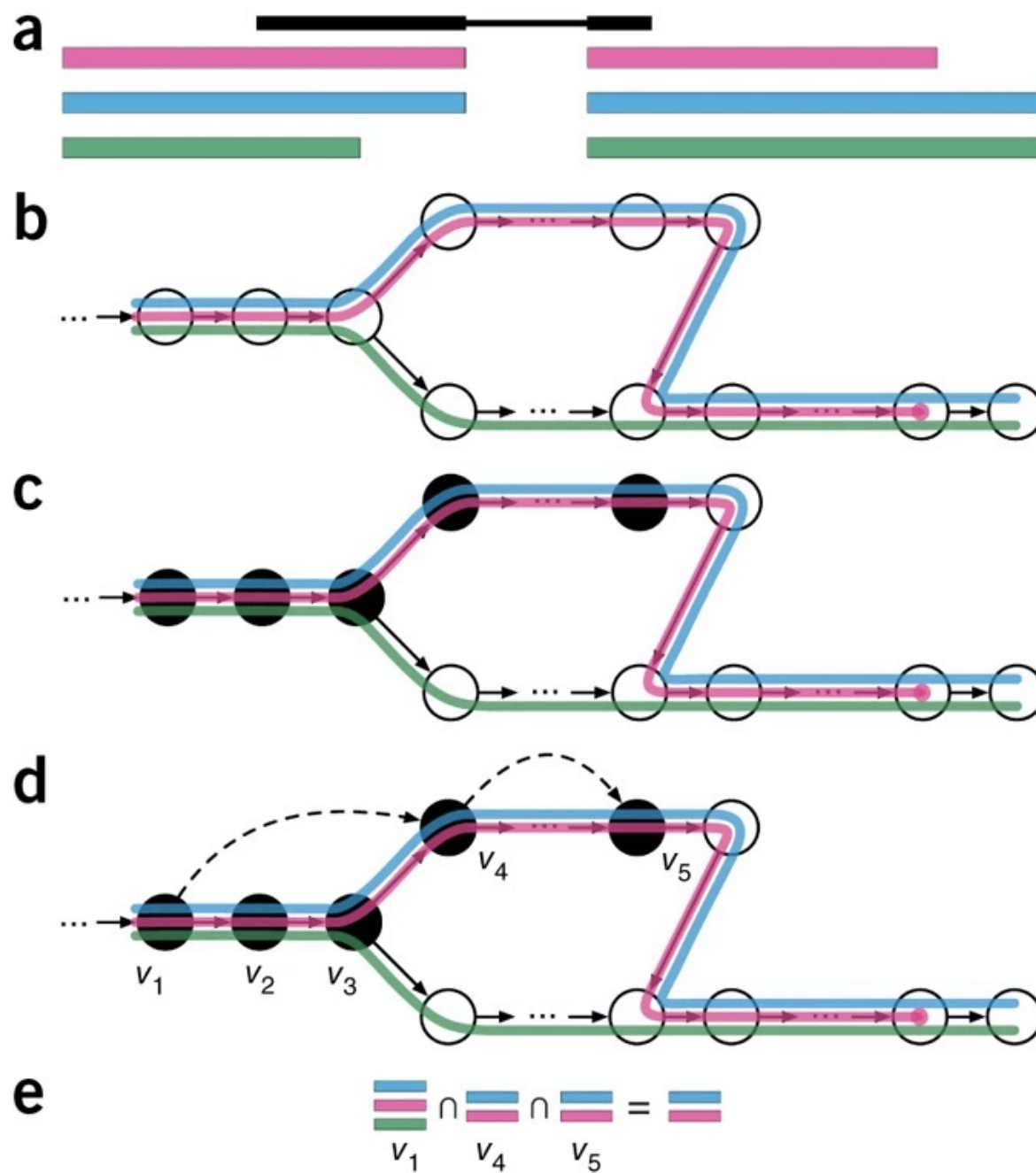
Pseudo-alignment and Quasi-mapping

- actually skips aligning to the genome
- takes transcriptome and splits it into “k-mers” of length $k = 31$ (so splits the transcriptome into chunks of 31-nt)
- builds an index from the k-mers for each transcript
- then, takes RNA-seq reads, makes k-mers from those, and figures out which transcripts those k-mers are consistent with
- pseudo-alignment: based on the number of k-mers that match, which transcript is most compatible with that read
- Is NOT aligning nucleotide sequences (so no SAM/BAM generated, just read counts per transcript)

a







There's no perfect algorithm for read alignment

BWA or Bowtie2 are still often used for whole-genome sequencing because it works well for variant calling

STAR/HISAT are common for RNA-seq when alignment visualization or splice junction analysis are needed

kallisto/Salmon are arguably the default choice for standard differential expression analysis, especially for large genomes

https://clstacy.github.io/GenomicDataAnalysis_Fa23/